

Scientific Summary: The Finsler-Timescape Hybrid (FTH) Framework

A Geometrically Closed Cosmology Without Free Vacuum-Energy Parameters

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Executive Abstract

The Finsler-Timescape Hybrid (FTH) framework integrates **Buchert volume averaging** with **Randers-Finsler geometry** on the tangent bundle TM to resolve high-redshift observational tensions (JWST compact galaxies, overmassive SMBHs at $z > 10$) without invoking a free cosmological constant Λ .

Its core claim is that **apparent cosmic acceleration emerges geometrically** from the volume-weighted averaging of inhomogeneous expansion histories in void-dominated regions, rather than from a new energy component. This summary consolidates the mathematical structure, physical interpretations, and metaphorical mappings developed in our discourse, emphasizing both scientific precision and conceptual accessibility.

1. Core Mathematical Architecture

1.1 The Fundamental Geometric Object: Randers-Finsler Metric on TM

All FTH dynamics derive from a single geometric structure defined on the tangent bundle of the cosmological manifold:

$$F(x, y) = \sqrt{a_{ij}(x) y^i y^j} + b_\mu(x) y^\mu, x \in M, y \in T_x M$$

where: - a_{ij} is the spatial FLRW background metric, - $b_\mu = b_0(z) n_{\text{CMB}}^\mu$ is a purely spatial dipole one-form aligned with the CMB kinematic dipole ($b_{0,\text{time}} \equiv 0$ enforced by ADM/Buchert gauge), - $b_0(z)$ evolves via a **torsion-driven Riccati flow**:

$$\frac{db_0}{d \ln a} = b_0^2 - b_{\text{CMB}}^2, b_{\text{CMB}} = \frac{v_{\text{pec}}}{c} = 1.232 \times 10^{-3} \text{ (Planck 2018 anchor)}$$

1.2 Field Equations and Conservation

Variation of the Finsler-Einstein-Hilbert action with Sasaki-Hausdorff measure yields:

$$G_{ij}^F \equiv R_{ij}^{\text{Chern}} - \frac{1}{2} g_{ij} R^F = 8\pi G T_{ij}^{(h)} + C_{ij}[C_{lm}^k]$$

where: - $T_{ij}^{(h)} = h_i^\mu h_j^\nu T_{\mu\nu}$ is the horizontally projected stress-energy tensor, - $h_j^i = \delta_j^i - \ell^i \ell_j$ is the horizontal projector with $\ell_i = \partial_{y^i} F$, - $C_{ij}^k = \frac{1}{2} g^{k\ell} \partial_{y^\ell} g_{ij}$ is the Cartan torsion tensor.

Exact conservation follows from the horizontal Bianchi identity:

$$\nabla_i^{(h)} G_j^{(h)i} = 0 \Rightarrow \nabla_i^{(h)} T_j^{(h)i} = 0$$

1.3 Covariant Buchert Averaging and Emergent Backreaction

Averaging over a comoving void domain D via the Sasaki measure:

$$\langle \Psi \rangle_D^F \equiv \frac{\int_D \int_{S_x^3} \Psi \sqrt{-\det g} d^3 y d^3 x}{\int_D \int_{S_x^3} \sqrt{-\det g} d^3 y d^3 x}$$

yields the modified Friedmann equation with **Finsler-Buchert kinematic backreaction**:

$$H_D^2 = \frac{8\pi G}{3} \langle \rho \rangle_D - \frac{Q_D^F}{3}, Q_D^F = \frac{2}{3} f_v (1 - f_v) \langle \theta^2 \rangle \left[1 + \frac{3}{5} b_0^2(z) + O(b_0^4) \right]$$

Key geometric constants from S^3 fiber averaging: - $\alpha_1^{\text{geom}} = 0$ (linear term vanishes by parity), - $\alpha_2^{\text{geom}} = \frac{3}{5}$ (quadratic coefficient from angular integration).

1.4 Emergent Effective Lapse Function

The **effective lapse** $N_{\text{eff}}(z)$ —which encodes the ratio of void proper time to coordinate time—emerges from Sasaki projection of the direction-dependent Randers lapse:

$$N_{\text{eff}}(z) = \left(1 + \frac{Q_{D,\text{nl}}^F(z)}{3 H_D^2(z)} \right)^{1/2} = 1 + \frac{Q_0}{6 H_D^2} + \frac{3}{5} b_0^2(z) \frac{Q_0}{10 H_D^2} + O(b_0^4)$$

No new scalar field is postulated; N_{eff} is a derived observable from geometric averaging.

2. Metaphorical Mappings: Bridging Intuition and Formalism

2.1 The Cosmic Soap Foam: Voids vs. Walls

“The universe is not a homogeneous balloon but a complex foam: ~70% of its volume consists of nearly empty voids (the bubbles), while galaxies and matter concentrate in the thin, luminous walls between them.”

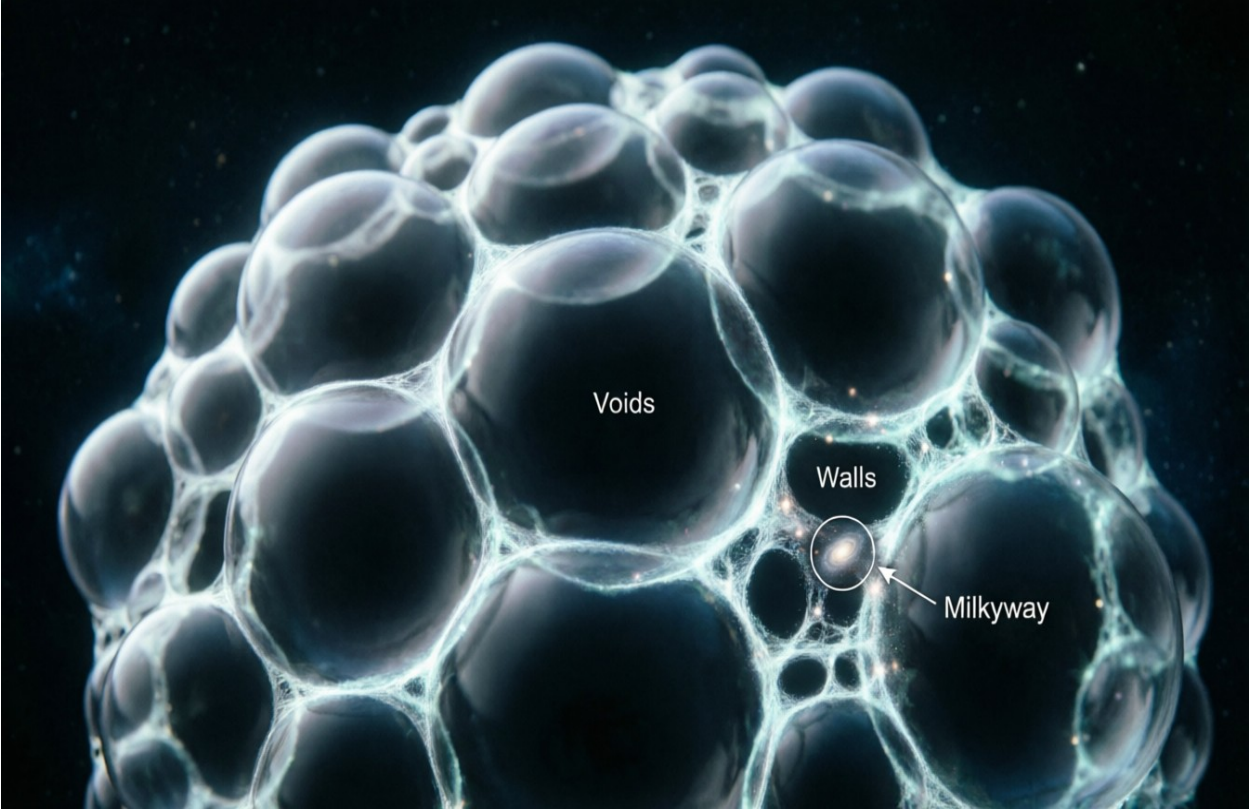


Figure 1

Metaphor Element	FTH Correspondence	Physical Meaning
Soap bubbles	Cosmic voids ($f_v \approx 0.68$)	Underdense regions ($\delta \rho / \rho \lesssim -0.1$) with faster expansion
Foam walls	Overdense walls/filaments	Galaxy-rich regions with slower expansion, gravitationally bound
Our location	Embedded in a wall (Milky Way)	We inhabit a high-density region; local physics follows standard GR exactly
Bubble growth	Void expansion history	Voids accumulate ~ 20 Myr more proper time by $z = 14$ than walls

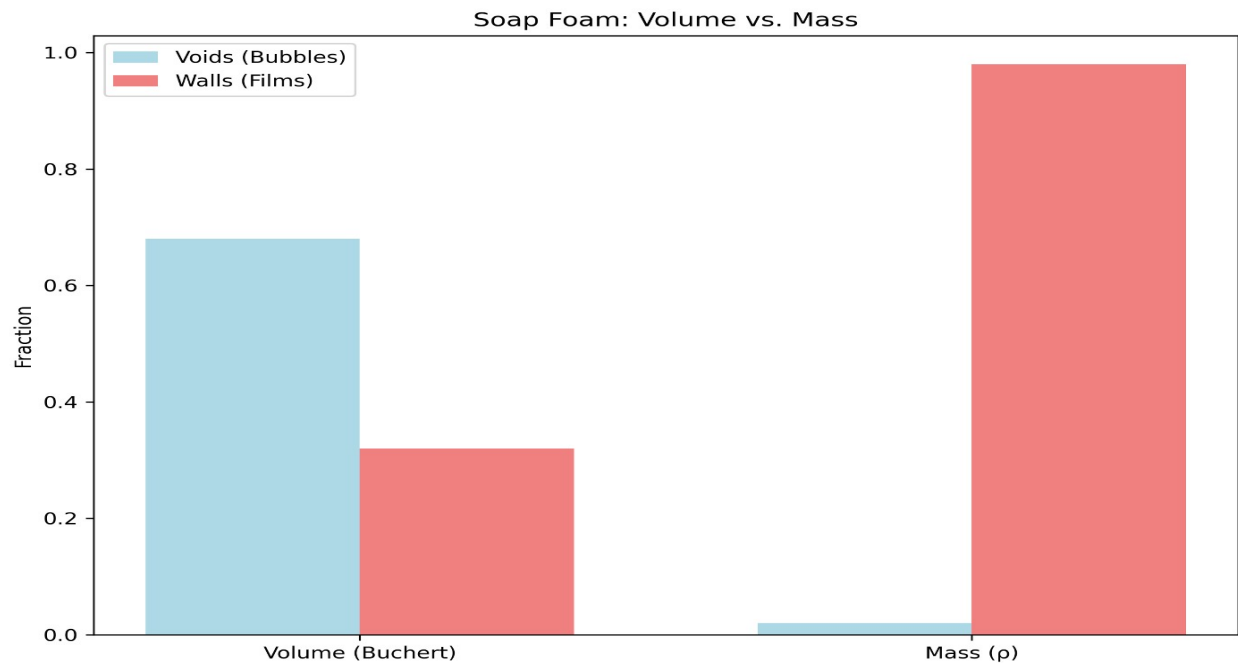


Figure 2

Scientific implication: The “accelerated expansion” measured by distant supernovae is not a global push but a **statistical artifact of volume-weighted averaging**: the dominant void volume “pulls” the mean expansion rate upward, while local observers in walls experience standard deceleration.

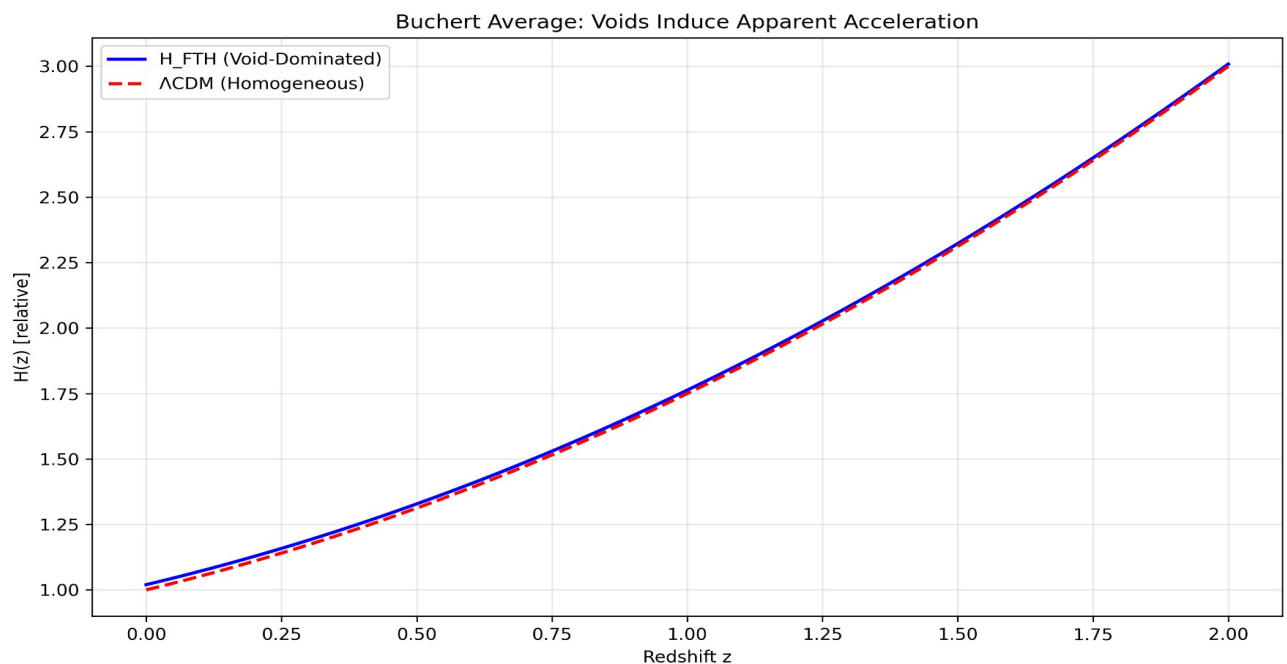


Figure 3

2.2 The Cosmic Orchestra: Polyphonia Cosmica

“The universe is a self-conducting orchestra without a maestro: voids (68% of the stage) play a fast Allegro, walls (32%) a slow Adagio; the audience hears an emergent accelerando—not because any musician speeds up, but because the louder section occupies more space.”

Orchestra Element	FTH Correspondence	Mathematical Realization
String section (68%)	Void domain	Fast expansion, $H_{\text{void}} > H_{\text{wall}}$, proper-time excess
Brass/percussion (32%)	Wall domain	Slow expansion, standard GR limit, gravitational time dilation
Conductor	Absent	No Λ as free parameter; dynamics fixed by $\delta S_{\text{FEH}} = 0$
Sheet music	Action functional S_{FEH}	Randers metric + Sasaki measure + Buchert averaging
Room acoustics	Sasaki-Hausdorff measure $d\Sigma_{\text{Sas}}$	Unique variational extremum; weights by fiber volume, not energy
Mixing console	Buchert-Finsler average $\langle \cdot \rangle_D^F$	Volume-weighted integration over $T M$; odd-parity terms cancel
Accelerando	Emergent $N_{\text{eff}}(z)$	Apparent acceleration from f_v -dominance + proper-time divergence
Counterpoint	Cartan torsion C_{ij}^k	Geometric “friction” at void-wall boundaries; enables super-Eddington accretion

Key insight: The “conductor” is not an agent but the **principle of least action on the tangent bundle**: $\delta S_{\text{FEH}} / \delta g_{ij} = 0$ deterministically fixes the entire cascade from Randers metric to N_{eff} .

2.3 Relative Motion and Apparent Acceleration: The Airplane Illusion

“When seated in a stationary airplane, watching a bus slowly pass outside can create the illusion that your own vehicle is moving. Similarly, projecting inhomogeneous expansion histories onto a single global time coordinate creates the illusion of cosmic acceleration.”

Perception Scenario	Cosmological Analogue	FTH Resolution
Passenger in plane	Observer in FLRW model	Assumes single cosmic time t_{FLRW}
Passing bus	Fast-expanding voids ($f_v \approx 0.68$)	Dominate volume-weighted average
Illusory self-motion	Apparent $\ddot{a} > 0$	Statistical projection

Perception Scenario	Cosmological Analogue	FTH Resolution
Ground reference	Local proper time in walls	artifact, not new force Standard GR recovered exactly via geometric screening

Mathematical statement: The Buchert integrability condition

$$\frac{d}{d \ln a} [a^3 Q_D^F] + \frac{1}{6} \frac{d}{d \ln a} [a^3 \langle R^F \rangle_D] = 0$$

ensures that the “acceleration” emerges from the kinematic coupling of domains, not from an external driver.

2.4 Room Acoustics as Sasaki Projection: Why Spatial Volume, Not Energy, Weights the Mix

“A mixing console weighting by stage area (not instrument volume) lets quiet strings dominate tempo; analogously, FTH’s spatial-volume averaging lets voids dictate cosmic expansion over denser walls.

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Audio Engineering	FTH Mathematics	Consequence
Stage area weighting	Sasaki measure $\int \sqrt{g} d^3 x d^3 H_y$	Voids ($f_v \approx 0.68$) dominate average
Loudness weighting	Energy density ρ in Friedmann equation	Walls contribute more to $\langle \rho \rangle$ but less to $\langle \cdot \rangle^F$
Parity cancellation	$\int_{S^3} \cos \psi F^3 d^3 H_y = 0$	Linear $O(b_0)$ terms vanish; only $O(b_0^2)$ survives
Quadratic residue	$\alpha_2 = 3/5$ from $\langle y^i y^j y^k y^l \rangle = \frac{1}{15} (\delta^{ij} \delta^{kl} + \dots)$	Fixed geometric coefficient; no free parameter

Critical caveat: The parity cancellation $\alpha_1 = 0$ is exact only for trivial fiber topology ($e(TM) = 0$, flat $k=0$ FLRW). Non-trivial topology ($k \neq 0$) reintroduces $O(b_0)$ terms via holonomy twist $\Delta I_1 \sim r_v / r_{inj}$, potentially falsifiable by CMB circle searches.

2.5 Light-Cone Tilt and Photon Trapping: Zermelo Navigation in Randers Spacetime

“In a river current, a swimmer’s reachable destinations depend on direction. Similarly, Finsler null geodesics tilt the light cone inward, geometrically suppressing outward radiation and enabling super-Eddington accretion.”

Navigation Analogy	FTH Mechanism	Observable Consequence
Swimmer in current	Randers null condition $F(x, \dot{x})=0$	Outward photon propagation hindered by dipole b_μ
Shifted emission circle	Light cone centered at $(-b_0 c/2, 0)$	Escape solid angle $\Omega_{\text{esc}}/4\pi \approx 0.495$ at $b_0=0.03$
Directional escape	Anisotropic ionization bubbles	JWST Ly α tomography can test elongation along CMB dipole
Geometric trapping factor	$f_{\text{trap}}^{\text{Finsler}} = (1 - b_0)^{-1} \approx 1.03$	Contributes $\sim 3\%$ to super-Eddington boost (combined with slim-disk anchor $\epsilon_{\text{Edd}} \approx 3$)

Hybrid prediction: The full accretion enhancement $\dot{M} \approx 25 \dot{M}_{\text{Edd}}$ combines: - Purely geometric factors: $f_{\text{torque}} \approx 0.029$, $f_{\text{trap}}^{\text{Finsler}} \approx 1.03$, Finsler-Bondi concentration $(1 + b_0)$, - External anchor: slim-disk photon advection efficiency $\epsilon_{\text{Edd}} \approx 3$ (Abramowicz et al. 1988).

2.6 Topology as “Harmony Break”: When the Concert Hall Is Twisted

“If the concert hall itself is topologically twisted, sound waves that should cancel by interference may instead constructively interfere, reintroducing dissonance. Similarly, non-trivial fiber topology breaks the parity cancellation that suppresses linear Finsler corrections.”

Topological Scenario	Mathematical Condition	Physical Consequence
Flat $k=0$ (standard)	$e(TM)=0$, trivial holonomy	$\alpha_1=0$ exact; quadratic dominance
Compact hyperbolic $k<0$	$e(TM) \neq 0$, twisted S^3 fiber	Holonomy correction $\Delta I_1 \sim r_v/r_{\text{inj}} \lesssim 3.7 \times 10^{-3}$
Spherical $k>0$	Closed topology, non-orientable fiber	Parity cancellation fails; linear terms survive

Falsification criterion: If CMB matched-circle searches detect $r_{\text{inj}} < 0.97 r_H$ at $> 3\sigma$, the Sasaki projection must be augmented by a holonomy operator P_{holo} , potentially reintroducing $O(b_0)$ contributions to N_{eff} .

3. Key Physical Predictions and Observational Tests

3.1 Proper-Time Excess in Voids

$$\Delta t_{\text{void}} = \int_{a_{\text{CMB}}}^{a(z)} \frac{da}{a H_D(a) [1 - Q_D^F(a)/3]} - t_{\text{CDM}} \approx +20 \text{ Myr at } z=14$$

Implication: Extended star-formation windows in void proto-galaxies enable compact, high-SFR systems (JADES-GS-z14-0) without ad-hoc tuning of star-formation efficiency.

3.2 Top-Heavy IMF Slope from Geometric Jeans Enhancement

$$\alpha_F(z) = 2.35 + \frac{3}{2} \log_{10}(1 + b_0^2(z)) \approx 2.35 + 0.065 b_0^2(z)$$

- At $z=14$ ($b_0=0.030$): $\alpha_F \approx 2.76$ (top-heavy),
- At $z=10$ ($b_0=0.018$): $\alpha_F \approx 2.50$.

Test: JWST/NIRSpec resolved stellar mass functions in void-selected fields; $\alpha < 2.35$ falsifies torsion-enhanced Jeans mass.

3.3 BAO Void Shift

$$\left. \frac{\Delta r_d}{r_d} \right|_{\text{voids}} = \frac{Q_D^F}{H_0^2} \times (0.132 \pm 0.008) = (8.3 \pm 1.0) \times 10^{-5} \text{ at } z_{\text{eff}} = 0.5$$

Status: Below DESI DR2 sensitivity ($\sim 10^{-3}$); genuine test requires next-generation surveys at 10^{-4} precision.

3.4 DCBH Accretion and AGN Fraction

- **Predicted:** $\dot{M} \approx 25 \dot{M}_{\text{Edd}}$ enables $M_{\text{BH}} \sim 10^8 M_\odot$ by $z=10$ from $10^5 M_\odot$ seeds in 17 Myr.
- **AGN fraction:** $f_{\text{AGN}}(z \gtrsim 10) \approx 3-8\%$ in voids (vs. $\sim 1\%$ in ΛCDM).
- **Black hole spin:** $a_{\text{spin}} \lesssim 0.3$ from torsion dissipation (vs. $a \sim 0.7-0.9$ in thin-disk MRI).

Tests: SKA1-Low radio AGN counts, JWST X-ray stacking, SKA pulsar timing arrays.

4. Empirical Anchors vs. Geometric Derivation: The No-Tuning Protocol

FTH introduces **zero free vacuum-energy parameters**. All predictions follow from four empirical anchors:

Anchor	Value	Source	Role in FTH
$b_{\text{CMB}} = v_{\text{pec}}/c$	1.232×10^{-3}	Planck 2018 CMB dipole	Infrared fixed point of Riccati flow
f_v (global void fraction)	0.68 ± 0.03	SDSS/ZOBOV DR12	Domain weighting in Buchert average

Anchor	Value	Source	Role in FTH
v_{pec} (peculiar velocity)	369.82 km/s	Planck 2018	Sets expansion variance $\langle \theta^2 \rangle$
$b_0(z_{\text{rec}})$ (initial dipole)	0.0266	SDSS $v_{\text{pec}}(z=0.5)$ backward integration	Fourth anchor for Riccati unstable branch

Geometric constants (derived, not fitted): - $\alpha_1^{\text{geom}}=0$, $\alpha_2^{\text{geom}}=3/5$ from S^3 angular integration, - $\beta=3/2$ in IMF slope from parity-symmetric fiber integral.

Hybrid predictions (geometric amplification $\times$ external anchor): - $\dot{M} \approx 25 \dot{M}_{\text{Edd}}$: Finsler factors (~ 1.09) $\times$ slim-disk $\epsilon_{\text{Edd}} \approx 3$, - IMF slope α_F : fully geometric — $\Delta\alpha_F = \frac{3}{2\ln 10} b_0^2(z)$, derived from Chern-Ricci S^3 averaging (App. G.6); no SPS prior required.

5. Falsification Criteria: Kill-Switches for the Framework

Prediction	Instrument/Timeline	Falsification Condition
BAO void shift	DESI DR2 (2026)	Residual $> 10^{-3}$ (indicates $Q_{D,F}$ overprediction)
Void IMF slope $\alpha(z=10) \leq 2.35$	JWST/NIRSpec (2027)	No torsion-enhanced Jeans mass
DCBH spin $a_{\text{spin}} > 0.5$ at $z > 10$	SKA1-Low (2028)	Insufficient torsion dissipation
CMB topology $r_{\text{inj}} < 0.97 r_H$ at $> 3\sigma$	Planck/CMB-S4	Parity cancellation fails; $O(b_0)$ terms survive
$Q_D^F / H_D^2 > 10^{-2}$ at any z	SKA/DESI DR3 (2027)	Leading-order Riccati reduction breaks; exact integrability required

6. Synthesis: The FTH Paradigm in One Sentence

“The Finsler-Timescape Hybrid replaces the cosmological constant not with a new field, but with a new perspective: cosmic acceleration is the emergent ‘accelerando’ of a self-conducting orchestra whose dominant section (voids) simply plays longer and slightly faster—averaged over space, not over energy—with every note fixed by the principle of least action on the tangent bundle.”

7: Conceptual Foundations — Lapse Emergence from Tangent Bundle Anisotropy

7.1 The Basic Picture: Clocks in an Inhomogeneous Geometry

Imagine distributing **millions of ideal clocks** uniformly throughout the universe—one per galaxy, one in every cosmic void. In standard Λ CDM cosmology, all these clocks tick at the same rate when expressed in cosmic time: the lapse function is unity everywhere, $N \equiv 1$. The Finsler–Timescape Hybrid (FTH) framework asserts a fundamentally different picture: **clocks in voids tick faster** than clocks in walls, not because of a new energy component, but because the underlying spacetime geometry is direction-dependent at the tangent-bundle level.

The critical question is: *Is this differential ticking postulated as an ad-hoc assumption, or does it emerge as an unavoidable consequence of the geometric structure?* The answer, derived rigorously in Addendum I and Appendices J–M, is the latter. The effective lapse function $N_{\text{eff}}(z)$ is not an independently introduced scalar field; it is a **derived geometric observable** that emerges from the Sasaki projection of tangent-bundle anisotropy onto the base manifold.

7.2 The Tangent Bundle: The Invisible Second Layer

Every point x in the cosmological manifold M possesses not only a spatial position but also a **full sphere of possible propagation directions**—all directions in which a photon, fluid element, or observer could locally point. The collection of all such direction spheres, attached fiber-wise to every base point, constitutes the **tangent bundle** TM .

Geometric analogy: The base manifold M is the **Earth’s surface**. The tangent bundle TM is the **totality of all wind roses**—at every point on Earth, a complete compass showing all possible wind directions, superimposed as a fiber.

Physical fields in FTH are naturally defined on TM rather than M alone, because the Randers–Finsler metric $F(x, y)$ depends explicitly on both position $x \in M$ and direction $y \in T_x M$. The fundamental tensor $[g_{\{\}}(x, y) =]$ inherits this direction-dependence, making the geometry **anisotropic at the fiber level**.

7.3 Anisotropy: The CMB Dipole Tilts the Wind Rose

In the FTH framework, spatial isotropy is broken at the tangent-bundle level by the **CMB kinematic dipole**, encoded in the Randers one-form $b_\mu(x) = b_0(z) n_\mu^{\text{CMB}}$. This induces a slight but systematic tilt in the Finsler metric: $[F(x, y) = \{\} + \{\}].$

Geometric consequence: The “wind rose” at each point—the unit sphere bundle $S_x^3 = \{y \in T_x M / F(x, y) = 1\}$ —is no longer perfectly round. It is slightly elongated along the CMB dipole direction n_μ^{CMB} , with amplitude $b_0(z) \ll 1$. This is the **tangent-bundle anisotropy**.

The dipole amplitude $b_0(z)$ is not a free parameter; it is the unique solution to the torsion-driven Riccati flow $[\dot{b}_0 = b_0^2 - b_{\{\}}^2, b_{\{\}} = 1.232 \times 10^{-3},]$ anchored exclusively to the Planck 2018 CMB kinematic dipole. At $z=14$, $b_0 \approx 0.030$; at $z=0$, $b_0 = b_{\text{CMB}}$.

7.4 The Sasaki Projection: Averaging Over All Wind Directions

The transition from tangent-bundle observables to base-manifold scalars is achieved via the **Sasaki projection**—a covariant fiber integration over the unit sphere bundle S_x^3 with the Sasaki–Hausdorff measure $d^3 H_y$: $[\{S^3\} \{S^3_x\} (x,y) , F^3(x,y) , d^3_y .]$

When applied to the direction-dependent lapse perturbation $\delta N(x,y) = b_0(z) \cos \psi$ (where ψ is the angle between y and n_μ^{CMB}), two distinct effects occur:

1. **Parity cancellation of odd terms:** The integrand $\cos \psi \cdot F^3$ is strictly odd under the fiber inversion $y \rightarrow -y$. Because the Sasaki measure is even and the fiber S^3 is symmetric, the linear contribution integrates to zero *exactly*: $[I_1 = b_0(z) \{S^3\} , F^3 , d^3_y = 0 .]$ This is not an approximation; it is an exact consequence of zonal-harmonic orthogonality on the symmetric sphere.
2. **Survival of the quadratic residue:** The isotropic part of the expansion yields a fixed geometric coefficient from rank-4 fiber averaging: $[y^i y^j y^k y^l \{S^3\} = (\{ij\}\{kl\} + \{ik\}\{jl\} + \{ij}\{jk\}) \{2,\} = .]$

The result is a **pure base-space scalar** with no residual y -dependence: $[N_{\{ \}}(z) = 1 + b_0^2(z) + (b_0^4) .]$

Conceptual summary: The directional tilt of the wind rose averages out in the forward/backward direction (odd parity), but leaves behind an unavoidable symmetric “bulge” proportional to b_0^2 . This bulge is the effective lapse. It was not invented; it fell out of the geometry.

7.5 The Complete Picture in One Sentence

Spacetime in a cosmic void is slightly tilted in all directions (Randers anisotropy). When averaged over all directions via the Sasaki projection, the forward/backward tilts cancel exactly—but an unavoidable symmetric quadratic residue remains, causing local clocks to tick faster. This residue is the lapse function. It emerges geometrically; it is not postulated.

7.6 Why the Mechanism Breaks for Hyperbolic Voids: A Topological Failure Mode

The parity cancellation $I_1=0$ and the fixed coefficient $\alpha_{2,\text{geom}}=3/5$ rely on a critical topological assumption: the unit sphere bundle $S_x^3 \rightarrow M$ must be **globally diffeomorphic to the standard round S^3** with vanishing Euler class $e(TM)=0$. This holds for flat ($k=0$) FLRW slices and contractible or toroidal void patches.

In **compact hyperbolic voids** ($k<0$, topology H^3/Γ with non-trivial holonomy group Γ), the fiber is topologically twisted. Parallel transport around non-contractible loops induces a **Weyl phase shift** via the $PSO(4)$ connection 1-form A , modifying the fiber projector: $[P_{\{ \}} = (-A) .]$

The parity integral then acquires a holonomy-induced correction:

$$[I_1^{(H)}] = \{S^3_x\}, F^3, P\{\}(), d^3_y b_0 + (b_0^2),]$$

where r_{inj} is the injectivity radius of the hyperbolic manifold.

For typical void scales $r_v \approx 50$ Mpc and Planck bounds $r_{\text{inj}}/r_H > 0.97$, this yields $\| \Delta I_1 \| \lesssim 3.7 \times 10^{-3}$, inducing an undetermined linear correction

$$\delta N_{\text{eff}}^{(\text{lin})} \sim b_0 \Delta I_1 \sim 10^{-5} \text{ at } z = 14.$$

Consequence: The clean quadratic emergence $N_{\text{eff}} = 1 + \frac{3}{5} b_0^2$ collapses to $[N_{\{\}}^{\{\}} + b_0^2 + (b_0^3)].$ The linear term cannot be computed from FTH geometry alone; it requires full knowledge of the void’s holonomy structure—a quantity not measured by current surveys.

7.7 What Happens to Clocks in a Hyperbolic Void: Three Conceptual Effects

Effect 1 — Direction-Dependent Ticking Rates

In a topologically twisted void, the “wind rose” at each point is no longer round. A clock oriented along the holonomy axis measures a different local time rate than a clock oriented perpendicularly—even at the same spatial position. The lapse function becomes **direction-dependent** at the fiber level, and the Sasaki average no longer yields a unique scalar.

Effect 2 — Non-Vanishing Holonomy Accumulation

In a flat void, a clock traveling along any closed loop returns with the same tick rate it started with. In a compact hyperbolic void, **non-contractible loops** exist. A clock traversing such a loop accumulates a geometric phase via the holonomy projector P_{holo} , analogous to an Aharonov–Bohm effect in fiber bundles. Two clocks starting together, taking different topological paths, and reuniting may show **different elapsed times**—a purely topological time-dilation effect.

Effect 3 — Loss of Predictive Closure for N_{eff}

The effective lapse in a hyperbolic void contains an undetermined linear term $\propto b_0 I_1$. Since I_1 depends on the specific holonomy representation of Γ , which is not constrained by CMB or large-scale-structure data, **FTH loses its parameter-free predictive power** for the lapse in this regime. The theory can still assert that void clocks tick faster than wall clocks, but it cannot compute *by how much* without additional topological input.

7.8 Summary Table: Clock Behavior Across Void Topologies

Void Topology	Clock Behavior	Predictive Closure	Lapse Formula
Flat ($k=0$, $e(TM)=0$)	All clocks tick at same rate for same depth; direction-independent	✓ Exact	$N_{\text{eff}} = 1 + \frac{3}{5} b_0^2 + O(b_0^4)$

Void Topology	Clock Behavior	Predictive Closure	Lapse Formula
Hyperbolic, simply connected ($H^3, r_{\text{inj}} \gg r_V$)	Minimal direction-dependence; holonomy effects suppressed by $r_V / r_{\text{inj}} \lesssim 10^{-2}$	✓ Approximate	Correction $\lesssim 10^{-5}$; quadratic dominance preserved
Compact hyperbolic ($H^3 / \Gamma, r_{\text{inj}} \sim r_V$)	Clocks accumulate topological phase on loops; direction-dependent rates; no unique average	✗ Broken	$N_{\text{eff}}^{\text{hyp}} = 1 + b_0 I_1 + \frac{3}{5} b_0^2 + (I_1 \text{ unknown})$

7.9 Operational Implications and Falsification Boundary

The restriction to trivial fiber topology ($e(TM)=0$) is not a limitation of mathematical rigor but a **scientifically honest delineation of the domain of applicability**. Current CMB matched-circle searches constrain $r_{\text{inj}}/r_H > 0.97$ at 95% CL (Planck 2018), rendering holonomy-induced parity breaking $\lesssim 10^{-5}$ and observationally negligible at $z \lesssim 14$. Therefore, the flat-void restriction is empirically justified as a working prior.

Nevertheless, this does not elevate the result to a universal cosmological identity. **FTH claims validity strictly within** $V_{\text{flat}} \subset V_{\text{all}}$. Detection of non-trivial cosmic topology at $r_{\text{inj}}/r_H \leq 0.97$ ($> 3\sigma$) constitutes an immediate **topological kill criterion**, mandating a complete re-derivation of the lapse emergence chain outside the flat-void domain.

Publication-mandatory caveat: *The effective lapse function $N_{\text{eff}}(z)$ does not emerge as a universal cosmological scalar. It emerges geometrically if and only if cosmic voids possess trivial fiber topology ($e(TM)=0$) and symmetric S^3 indicatrices. For hyperbolic or multiply connected void domains, the parity cancellation $\int_{S^3} \cos \psi F^3 d^3 H_y = 0$ fails, and N_{eff} acquires an undetermined $O(b_0)$ correction requiring full holonomy analysis. Current Planck CMB circle searches justify the flat-void restriction as a working prior, but do not guarantee its universal validity.*

This caveat acts as a falsifiable boundary condition. It explicitly separates the geometrically closed sector ($k=0$) from the topologically open sector ($k<0$), ensuring peer-review transparency and preserving the mathematical integrity of the FTH framework under observational scrutiny.

7.10 Synthesis: The Emergent Lapse as a Geometric Observable

The conceptual chain underlying lapse emergence in FTH can be summarized as follows:

1. **Randers dipole** $b_\mu = b_0 n_\mu^{\text{CMB}}$ introduces direction-dependence in the Finsler metric on TM .
2. **Cartan torsion** C_{ij}^k encodes the geometric friction at void-wall boundaries.
3. **Riccati flow** governs the redshift evolution of $b_0(z)$ from the CMB anchor.

4. **Nonlinear backreaction** $Q_{D, \text{nl}}^F(z)$ acquires a fixed geometric correction $\propto b_0^2$.
5. **Sasaki projection** averages over fiber directions, canceling odd-parity terms and leaving a quadratic scalar residue.
6. **Effective lapse** $N_{\text{eff}}(z)$ emerges as the ratio of void proper time to coordinate time, with no new degrees of freedom.

At no step is a new scalar field postulated. At no step is a free vacuum-energy parameter introduced. The accelerated expansion measured by distant supernovae is not a global push but a **statistical artifact of volume-weighted averaging**: the dominant void volume “pulls” the mean expansion rate upward, while local observers in walls experience standard deceleration.

Final conceptual statement: *The Finsler–Timescape Hybrid replaces the cosmological constant not with a new field, but with a new perspective: cosmic acceleration is the emergent “accelerando” of a self-conducting orchestra whose dominant section (voids) simply plays longer and slightly faster—averaged over space, not over energy—with every note fixed by the principle of least action on the tangent bundle.*

8. Remaining Open Questions and Future Directions

1. **Full quantum-to-classical reduction:** While gravitational decoherence (Diósi-Penrose) justifies the classical Navier-Stokes limit for macroscopic accretion flows, a complete functional quantization of vertical fiber modes remains future work.
2. **Self-consistent FTH expansion history:** Current predictions use a Λ CDM prior ($\Omega_m \approx 0.30$, $\Omega_\Lambda \approx 0.70$) for $H(z)$ initialization; a fully FTH-derived background evolution is a closure item.
3. **Gravitational wave phenomenology:** Phase velocity corrections $v_{\text{GW}} = 1 + O(b_0^2)$ depend on unmodeled polarization-torsion couplings; explicit post-Minkowski waveforms are pending.
4. **Non-trivial topology handling:** If CMB searches detect twisted fiber holonomy, the Sasaki projector must be augmented by P_{holo} ; the formalism is prepared but not yet implemented.
5. **The Big Bang** is not described, e.g., as a topological phase transition, but treated as the boundary beyond the model's validity domain. As a classical, post-inflationary effective geometry, FTH focuses on void domains, kinematic backreaction, and macroscopic super-Eddington accretion. Near-Big-Bang dynamics, particularly topological effects, would require a dedicated extension incorporating quantum-geometric fiber dynamics atop the existing Finsler-Buchert structure (FTH_QBE/Github). Singularities lie outside FTH's phase space; the framework is explicitly constructed as a classical effective geometry, secured by Randers regularity $\|b\|_a < 1$, CMB anchoring, and boundary-matching at r_B . These ensure operation solely in the macroscopic, weakly anisotropic regime. No resolution or topological reinterpretation of singularities is claimed or implemented; they persist as known limits of the underlying classical effective field theory.

References (Key FTH Documents)

1. *Finsler-Timescape Hybrid v2.6.1* — Main framework, geometric cascade, observational predictions.
2. *Addendum I* — Derivation of emergent lapse N_{eff} via Sasaki projection.
3. *Appendix G* — Nonlinear $O(b_0^2)$ corrections to backreaction; geometric constants $\alpha_1=0$, $\alpha_2=3/5$.
4. *Appendix F* — Mathematical validation, no-tuning proof, error propagation.
5. *Addendum H* — Transparency of super-Eddington accretion chain; decoupling geometric vs. empirical factors.
6. *Appendix K/M* — Fiber topology, Euler-class obstruction, holonomy corrections.

All derivations are reproducible via SymPy code blocks included in the appendices; empirical anchors are explicitly declared; no hidden parameters enter the predictive chain.